

Advanced Information Product Manual



~~Programme Specification~~

~~Title of Course: General Law Masters LLM (Full Time or Part Time)~~

~~Date Specification Produced: November 2018~~

~~Date Specification Last Revised: August 2023~~

PROGRAMME SPECIFICATION

This Programme Specification is designed for prospective students, current students, academic staff and potential employers. It provides a concise summary of the main features of the programme and the intended learning outcomes that a typical student might reasonably be expected to achieve and demonstrate if he/she takes full advantage of the learning opportunities that are provided. More detailed information on the teaching, learning and assessment methods, learning outcomes and content of each module can be found in Student Handbooks and Module Descriptors.

Examples of completed programme specifications can be found on the KU Programme Specification Archive

SECTION 1: GENERAL INFORMATION

<u>Engineering Specification Title:</u>	LLM
<u>Advanced Information Product Manual</u>	
<u>Awarding Institution:</u>	Kingston University

Teaching Institution: Kingston University

Location: Kingston School of Law, Social and Behavioural Sciences,
Faculty of Business and Social Sciences, Penrhyn Road

Programme Accredited by:

SECTION 2: THE PROGRAMME

A. Programme Introduction

The essence of the LLM is that, within an advanced academic environment, students will have the opportunity to engage with a range of legal issues which may be national or international, within their theoretical, social and economic contexts.

The LLM is designed for students who want maximum flexibility. The programme allows students to tailor their studies to their aspirations by choosing four elective modules from a range of specialist subjects, such as Arbitration, Alternative Dispute Resolution, Company Law, Employment Law, Evidence and Arbitral Awards, Law of International Trade, Legal Aspects of Business Finance and Insolvency, Law of Technology and Innovation, Family Law, Human Rights in Context and Theoretical Perspectives on Contemporary Legal Environments. The Legal Research and Dissertation module allows students to undertake a proper academic research project in any legal field of their interest—even beyond the taught modules—under the expert supervision of one of our scholars.

Students can choose between the full-time or the part-time programme. The full-time programme lasts one year and students are required to take two modules in the first term and three in the second (one of which is Legal Research and Dissertation). The part-time programme lasts two years and students are required to take one module for every term of each year plus Legal Research and Dissertation in the second term of the second year.

The LLM is not limited to traditional academic learning. Students on the LLM programme have the opportunity to take part in extracurricular activities such as mooting, the development of mediation skills and advocacy particularly through involvement in the Kingston Mooting and Advocacy Society. These activities help to develop presentational skills and interpersonal skills. Students will be also offered the opportunity, also through involvement in the Kingston

~~Law Society, to engage with the legal professions and industry—for instance, through court visits and networking and social events with solicitors, barristers and other external stakeholders.~~

~~The programme aims at offering a number of occasions to engage with Kingston University's research environment—for instance, by being involved in the activities of research centres and groups. In particular, students also have the opportunity to participate in the regular research seminars that are held throughout the teaching terms. These seminars, in which members of staff present their research interests, are interactive and allow students and staff to develop points of interest.~~

~~After completion of the LLM, many students return to practise in their chosen field, having a deeper knowledge of a specialist area, while others embark on a new career, having found a new interest after studying a new subject.~~

~~B. Aims of the Programme~~

~~The main aims of the LLM are to provide students with the opportunity to:~~

- ~~• take advantage of a very flexible programme of taught modules by creating a programme of study designed to suit their own interests and aspirations~~
- ~~• acquire a detailed knowledge and understanding of, and develop the ability to analyse, evaluate and reflect upon, legal rules relating to areas selected by them~~
- ~~• apply law to complex situations~~
- ~~• develop research and investigative skills and a critical and research-oriented approach to the study of areas of law in which they are interested~~
- ~~• develop their interest at postgraduate level of related areas of legal study~~
- ~~• enhance their career opportunities with a postgraduate qualification which emphasises the skills which are central to work success in a knowledge economy~~
- ~~• develop their lifelong learning and continuing professional development.~~

~~C. Intended Learning Outcomes~~

~~The programme provides opportunities for students to develop and demonstrate legal knowledge and understanding, legal skills and other attributes in the following areas. The programme outcomes are referenced to the Framework for Higher Education Qualifications in England, Wales and Northern Ireland, and relate to the typical student.~~

PROGRAMME SPECIFICATION

<u>Document No.</u> Programme Learning Outcomes		<u>SPEC-KA210A</u>			
<u>Version</u>	1.1 Knowledge and Understanding On completion of the course students will be able to:		Intellectual Skills On completion of the course students will be able to		Subject Practical Skills On completion of the course students will be able to
<u>Description</u> A1	<u>This document mainly introduces the new generation Kaadas Door Lock. The content mainly includes its interfaces, accessories, features, specifications, quick start, and software function definition.</u> <u>This device must be used in conjunction</u>	B1	Submit areas of law studied to critical analysis and evaluation	G1	Communicate the results of independent legal research both in writing and, if appropriate to any particular taught law modules, in other ways

PROGRAMME SPECIFICATION

	<u>with a security enabled Z-Wave™ controller in order to fully utilize all implemented functions. This product can be operated in any Z-Wave™ network with other Z-Wave™ certified devices from other manufacturers. All non-battery operated nodes within the network will act as repeaters regardless of vendor to increase reliability of the network.</u> Demonstrate an in-depth knowledge and understanding of the sources of law, theory and practice				
--	--	--	--	--	--

PROGRAMME SPECIFICATION

	in the relevant area of law				
Written By A2	Critically appreciate the national or international social and economic context of the law in the relevant area	B2	Analyse legal problems and issues, taking due account of any incompleteness of data or information, and arrive at well-reasoned and supportable legal conclusions	C2	Use appropriate ICT to present legal text and data
Date A3	Evaluate the effectiveness of legal policy and practice developments in responding to current issues affecting the relevant area of law	B3	Carry out independent data collection and synthesise it so as to resolve legal problems/issues	C3	Search for and retrieve information from ICT sources relevant for law

1.

Ln

PROGRAMME SPECIFICATION

In addition to the programme learning outcomes identified overleaf, the programme of study defined in this programme specification will allow students to develop a range of Key Skills as follows:

Key Skills						
Self-Awareness Skills	Communication Skills	Interpersonal Skills	Research and information Literacy Skills	Numeracy Skills	Management & Leadership Skills	Creativity and Problem-Solving Skills
Take responsibility for own learning and plan for and record own personal development	Express ideas clearly and unambiguously in writing and the spoken work	Work well with others in a group or team	Search for and select relevant sources of information	Collect data from primary and secondary sources and use appropriate methods to manipulate and analyse this data	Determine the scope of a task (or project)	Apply scientific and other knowledge to analyse and evaluate information and data and to find solutions to problems
Recognise own academic strengths and weaknesses, reflect on performance and progress and respond to feedback	Present, challenge and defend ideas and results effectively orally and in writing	Work flexibly and respond to change	Critically evaluate information and use it appropriately	Present and record data in appropriate formats	Identify resources needed to undertake the task (or project) and to schedule and manage the resources	Work with complex ideas and justify judgements made through effective use of evidence
Organise self effectively, agreeing and setting realistic targets, accessing	Actively listen and respond appropriately to ideas of others	Discuss and debate with others and make concession to reach agreement	Apply the ethical and legal requirements in both the access and use of information	Interpret and evaluate data to inform and justify arguments	Evidence ability to successfully complete and evaluate a task (or project), revising	

PROGRAMME SPECIFICATION

support where appropriate and managing time to achieve targets					the plan where necessary	
<u>V</u> <u>e</u> <u>r</u> <u>s</u> <u>i</u> <u>o</u> <u>n</u> W e r k e ff e ti v e ly w it h	Give, accept and respond to constructive feedback	Accurately cite and reference information sources	<u>Brief description of changes</u> Be aware of issues of selection, accuracy and uncertainty in the collection and analysis of data	Motivate and direct others to enable an effective contribution from all participants		

ii	
iii	
iv	
v	
vi	
vii	
viii	
ix	
x	
xi	
xii	
xiii	
xiv	
xv	
xvi	
xvii	
xviii	
xix	
xx	
xxi	
xxii	
xxiii	
xxiv	
xxv	
xxvi	
xxvii	
xxviii	
xxix	
xxx	

PROGRAMME SPECIFICATION

t e x t s						
1 : 0 8	202 3.0 9.2 8	First revision. Show sensitivity and respect for diverse values and beliefs	Use software and IT technology as appropriate			

D. Entry Requirements

The following categories of applicant will normally meet the entry requirements for the LLM:

- applicants with a good honours degree in law (class 2(ii) and above) or non-UK equivalent
- applicants with a good joint honours degree (class 2(ii) and above) which comprises the study of law as a major component (for example, Accounting and Law, Combined Studies)
- applicants who have successfully completed a Conversion Graduate Diploma, a Legal Practice Course, a Bar Vocational Course, or non-UK equivalent
- solicitors or barristers

We welcome applications from international students however those students whose first language is not English should have one of the following recognised qualifications:

- IELTS (British Council) test score of 6.5 with no element less than 6.0
- TOEFL iBT 88 overall with R-18, L-17, S-20, W-17

English language support is offered both before and after enrolment. Students have an English test during induction week and are advised as to whether they need English language support classes during the LLM programme. English language support classes are offered in each semester on the LLM.

E. Programme Structure

This programme is offered in full-time and part-time modes with intake in September only, and leads to the award of LLM.

E1. Professional and Statutory Regulatory Bodies**E2. Work-based learning, including sandwich programmes**

N/A

E3. Outline Programme Structure

The programme consists of five modules: four at 30 credit points and one at 60 credit points. A student must complete 180 credits for a Masters degree.

The LLM is designed for students who want maximum flexibility: students will choose four taught modules from any of the electives, and will undertake a dissertation in an area of their choice, subject to the availability of an appropriate supervisor.

All students will be provided with the University regulations. Full details of each module are provided in the module descriptors and student module handbooks.

|

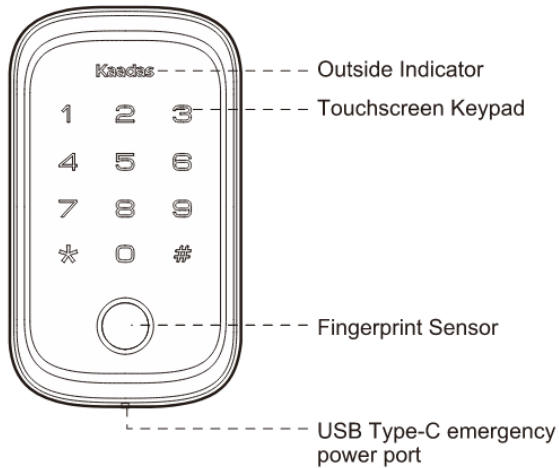
Table of Content1: Modules

1	INTERFACES & ACCESSORIES.....	18
2	FEATURES & SPECIFICATIONS.....	21
2.1	Structural Characteristics.....	21
2.2	Hardware Characteristics.....	24
2.3	Software Characteristics.....	1
3	PRODUCT QUICK START	3
3.1	Important safety information.....	3
3.2	About Z-Wave™.....	3
3.3	About SmartStart.....	3
3.4	How to add the product into Z-Wave™ network.....	3
3.5	How to remove the product from Z-Wave™ network.....	3
3.6	How to factory reset.....	4
3.7	Z-Wave™ DSK Location.....	4
3.8	About Product	4
3.9	About Security.....	4
4	SOFTWARE FUNCTION DEFINITION.....	5
4.1	User Behavior Interaction.....	5
4.2	Supported Command Classes.....	6
4.3	Basic Command Class mapping.....	6
4.4	ZWAVEPLUS INFO.....	7
4.5	Manufacturer Specific.....	7
4.6	Version	7
4.7	Association Group Info.....	8
4.8	Notification	9
4.9	Door Lock	10
4.10	Door Lock Logging.....	10
4.11	User Code	10
4.12	Schedule	11
4.13	Battery	11

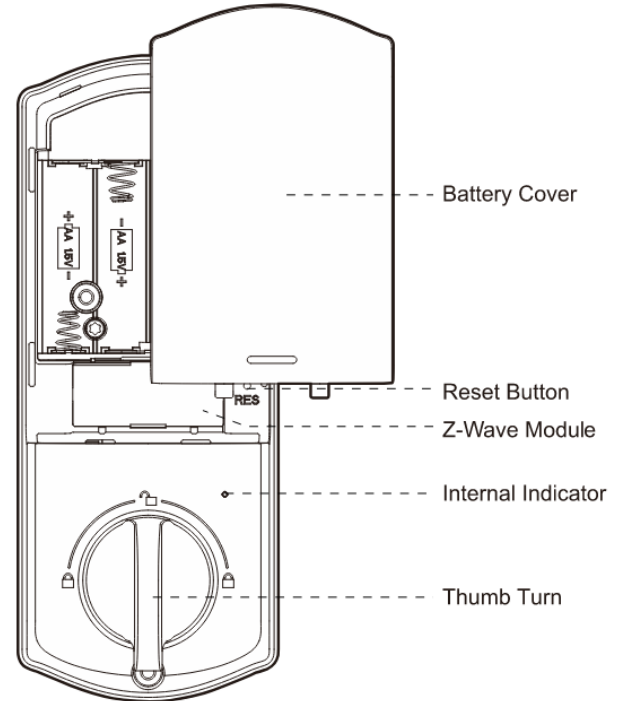
4.14	Indicator	11
4.15	Time	12
4.16	Time Parameters.....	12
4.17	Configuration	12

INTERFACES & ACCESSORIES

Exterior Assembly



Interior Assembly



Terminology		Level 7				Description	
Compulsory module		Module code	Credit Value	Level	Teaching Block		
Legal Research and Dissertation		LL7080	60	7	2		
Elective modules		Module code	Credit Value	Level	Teaching Block		
Alternative Dispute Resolution		LL7033	30	7	4		
Arbitration		LL7034	30	7	4		
Company Law		LL7035	30	7	4		
Employment Law		LL7045	30	7	4		
Evidence and Arbitral Awards		LL7042	30	7	2		
Law of International Trade		LL7039	30	7	2		
Legal Aspects of Business Finance and Insolvency Law		LL7086	30	7	2		
Touchscreen Keypad	Used to operate various functions of the door	LL7038	30	7	4 or 2		

	lock Labour Law in the Global Economy				
Thumb Turn Human Rights in Context	For manual lock / unlock operation LL7001	30	7	1 or 2	
Reset button Family Law	Restore factory settings LL7003	30	7	1 or 2	
Law of Technology and Innovation	LL7004	30	7	1 or 2	
Theoretical Perspectives on Contemporary Legal Environments	LL7002	30	7	1 or 2	
Energy and Climate Change Law	LL7056	30	7	1 or 2	

- ~~Students exiting the programme with 60 credits are eligible for the award of PgCert~~
- ~~Students exiting the programme with 120 credits are eligible for the award of PgDip~~

~~F. Principles of Teaching, Learning and Assessment~~

~~There are essentially two elements of the LLM, the taught modules and the Dissertation stage. The primary aim of the elements of teaching and learning on the LLM is to enable the programme aims, learning outcomes and the aims and learning outcomes of individual modules, both taught modules and the Legal Research and Dissertation, to be achieved by the student. In general terms, the teaching and learning philosophy of the continuous assessment programme is to enable the students to demonstrate their ability to study independently and it is envisaged that a relatively small proportion of student learning time will be staff student contact time.~~

~~The majority of the learning time of the student in relation to the taught modules will be spent in independent study, consisting of:~~

- ~~directed learning, giving the student specific tasks in preparation for, and in support of, class-based programmes, for example, the preparation of oral or written presentations, case analysis etc;~~
- ~~private learning time, comprising the preparation of material for seminars and workshops, reviewing of ideas introduced during contact time and directed research and development through reading, using recommended reading as a starting point;~~
- ~~preparation of assessed work.~~

~~Student learning is supported by the Learning Resources Centre law collection, by law specific Web-based resources such as LexisNexis and WestLaw and by the university online learning management system, Canvas. Student learning is further enhanced by the opportunity to take part in skills competitions, e.g. the National Mediation Competition and the National Negotiation Competition.~~

For the Legal Research and Dissertation, seminars will combine formal presentation of ideas and issues with exercises. Students will be encouraged to decide on their dissertation area at the earliest possible stage, and wherever possible, exercises will be aimed at helping them to develop ideas relevant to their dissertation research. Each student will be provided with a Research Handbook setting out in detail the various aspects of the Legal Research and Dissertation module.

Once their research topic has been approved, students will be allocated a personal supervisor, to whom they will submit their proposal for approval. It will be stressed to students, both during the seminars and in the Research Handbook, that the Dissertation is their individual work but each student is entitled to up to 3 hours of face-to-face supervision time with his/her supervisor to address problems that they might encounter.

The assessments are planned to reflect the aims and to achieve the learning outcomes of the programme. They are designed to monitor student performance regularly in the context of specific requirements of individual modules, and to ultimately assess the knowledge and skills essential to meet the overall requirements of the Master's degree. The assessment tutor schedules the coursework to ensure an even distribution of workload for students on the programme.

G. Support for Students and their Learning

Students are supported by:

- Course and module handbooks
- A Course Director
- Module leader for each module
- A designated programme administrator
- Each student is supported throughout the programme by a personal tutor. The personal tutor is available to discuss the student's academic studies and monitor progress throughout the programme.
- An induction session at the beginning of the programme
- The Centre for Academic Support and Enhancement (CASE) supports students in the School of Arts, Culture and Communication (Kingston School of Art) and the School of Law, Social and Behavioural Sciences (Faculty of Business and Social Sciences) with structuring and editing traditional academic writing.
- Staff Student Consultative Committee
- Technical support to advise students on IT and the use of software
- Canvas—a versatile on-line interactive intranet and learning environment
- Student support facilities that provide advice on issues such as finance, regulations, legal matters, accommodation, international student support etc.
- Disabled student support
- The Students' Union
- Careers and Employability Service

H. Ensuring and Enhancing the Quality of the Course

~~The~~ FEATURES & SPECIFICATIONS

Structural Characteristics

~~University has several methods for evaluating and improving the quality and standards of its provision. These include:~~

- ~~• External examiners~~
- ~~• Boards of study with student representation~~
- ~~• Annual review and development~~
- ~~• Periodic review undertaken at the subject level~~
- ~~• Student evaluation~~
- ~~• Moderation policies~~

~~I.—Employability Statement~~

~~Students who successfully complete the LLM will have acquired significant research and analytical skills in their chosen areas of law and should therefore have substantially enhanced their prospects of gaining employment as, for example, professional or in-house legal advisors and in other areas of work where the ability to apply and advise upon legal issues is significant.~~

~~J.—Approved Variants from the Undergraduate and Postgraduate Regulations~~

None

~~K.—Other sources of information that you may wish to consult~~

~~All students will be provided with the University regulations. Details about the programme will be provided in the course handbook, and full details of each module will be provided in module handbooks.~~

Development of Programme Learning Outcomes in Modules

This map identifies where the programme learning outcomes are summatively assessed across the modules for this programme. It provides an aid to academic staff in understanding how individual modules contribute to the programme aims, a means to help students monitor their own learning, personal and professional development as the programme progresses and a checklist for quality assurance purposes. Include both core and option modules.

[illegible]

	A3	S	S	S	S	S	S	S	S	S	S	S	S
Intellectual Skills	B1	S	S	S	S	S	S	S	S	S	S	S	S
	B2	S	S	S	S	S	S	S	S	S	S	S	S
	B3	S	S	S	S	S	S	S	S	S	S	S	S
Practical Skills	C1	S	S	S	S	S	S	S	S	S	S	S	S
	C2	S	S	S	S	S	S	S	S	S	S	S	S

Hardware Characteristics

~~S indicates where a summative assessment occurs.~~

~~Students will be provided with formative assessment opportunities throughout the course to practice and develop their proficiency in the range of assessment methods utilised.~~

Technical Annex

<u>Parameter</u> Final Award(s):	<u>Value</u> LLM
<u>Module</u> Intermediate Award(s):	ZGM130S037HGN1PgDip in Law PgCort in Law
<u>TX Power</u> Minimum period of registration:	Max: 13dBm Full time 1 year; part time 2 years
<u>Antenna</u> <u>Distance</u> Maximum period of registration:	40m (Indoor) /150m (Outdoor) Full time 3 years; part time 6 years
<u>Battery Info</u> FHEQ Level for the Final Award:	AA alkaline battery x4Masters
<u>Working Current</u> QAA Subject Benchmark:	Module: 10mA(Wake Up), 40mA (Send message) Door Lock body: 52mA(Wake Up), 190mA (operate door lock)
<u>Standby Current</u> Modes of Delivery:	Module: 24uA Door Lock body: 170uA Full time; part time

Language of Delivery:

English

Faculty:

Faculty of Business and Social Sciences

School:

School of Law, Social and Behavioural Sciences

JACS code:

M100

~~This is the Joint Academic Coding System (JACS) agreed jointly by UCAS and HESA.~~

UCAS Code:**Course Code:**

See overleaf

Route Code

See overleaf

Software Characteristics

<u>Parameter</u>	<u>Value</u>
------------------	--------------

<u>Wireless Technology</u>	<u>Z-Wave™</u>	
<u>Certification Type</u>	<u>Z-Wave Plus™ V2</u>	
<u>SDK Version</u>	<u>7.18.1</u>	
<u>Library Type</u>	<u>Enhanced 232 Slave</u>	
<u>Role Type</u>	<u>ZWAVEPLUS_INFO_REPORT_ROLE_TYPE_SLAVE_SLEEPING_LISTENING (0x07)</u>	
<u>Generic Device Type</u>	<u>GENERIC_TYPE_ENTRY_CONTROL (0x40)</u>	
<u>Specific Device Type</u>	<u>SPECIFIC_TYPE_DOOR_LOCK (0x01)</u>	
<u>Security Class</u>	<u>Non-Security, S0, S2-ACCESS</u>	
<u>SmartStart</u>	<u>Support. After powering on, SmartStart is auto active if it's out of the network.</u>	
<u>Over The Air (OTA)</u>	<u>Support. Firmware can be updated via RF.</u>	
<u>Multichannel Device</u>	<u>No</u>	
<u>Association</u>	<u>Support. Refer to Section 4.7 Association Group Info.</u>	
<u>Factory Reset</u>	<u>Support. Refer to Section 3.6 How to factory reset.</u>	
<u>Power-down Memory</u>	<u>Support. All command settings will stay unchanged even power down.</u>	
<u>Timed battery report</u>	<u>Support. Refer to Configuration Parameter 3.</u>	
<u>Low battery warning</u>	<u>Support.</u>	
<u>Door State Report</u>	<u>Support. When door lock mode changed, send out notification via Group 1.</u>	
<u>User code Report</u>	<u>Support. When user code changed, send out notification via Group 1.</u>	Route-Code
Programme	Course-code	

LLM-Law	PPLAW1LAW01	PPLAW1LAW01
LLM-Law 1YR-FT	PFLAW1LAW01	PFLAW1LAW01

PRODUCT QUICK START

Important safety information

Please read this Engineering Specification carefully for correct and effective use.

Failure to follow the recommendations set forth by Kaadas Limited may be dangerous or cause a violation of the law. The manufacturer, importer, distributor, and/or reseller will not be held responsible for any loss or damage resulting from not following any instruction in this guide or in other materials.

The product is intended for indoor use in dry locations only. Do not use in damp, moist, and /or wet locations.

About Z-Wave™

Z-Wave™ is the international wireless protocol for communication in the Smart Home.

Z-Wave™ ensures a reliable communication by reconfirming every message (two-way communication) and every mains powered node can act as a repeater for other nodes (meshed network) in case the receiver is not in direct wireless range of the transmitter.

This device and every other certified Z-Wave™ device can be used together with any other certified Z-Wave™ device regardless of brand and origin as long as both are suited for the same frequency range.

If a device supports secure communication it will communicate with other devices secure as long as this device provides the same or a higher level of security. Otherwise it will automatically turn into a lower level of security to maintain backward compatibility.

About SmartStart

SmartStart enabled products can be added into a Z-Wave™ network by scanning the Z-Wave™ QR Code present on the product with a controller providing SmartStart inclusion. No further action is required and the SmartStart product will be added automatically within 10 minutes of being switched on in the network vicinity.

How to add the product into Z-Wave™ network

1. Follow the user guide of hub to enter inclusion mode.
2. Operate on lock following guide below:
 - 1) Enter master mode(* + * + Master PIN Code + #)
 - 2) Input "0" to enter "Network settings"
 - 3) Input "1 + #" to enter "Include mode"

How to remove the product from Z-Wave™ network

1. Follow the user guide of hub to enter exclusion mode.
2. Operate on lock following guide below:
 - 1) Enter master mode(* + * + Master PIN Code + #)
 - 2) Input "0" to enter "Network settings"
 - 3) Input "2 + #" to enter "Exclusion mode"

How to factory reset

1. Operations on lock

- 1) Keep the door open and the lock unlocked (thumb turn will be vertical).
- 2) Using the Reset Tool, press and hold the Reset Button for 5 seconds until you hear the "BEEP" and the latch rotates, turning the thumb turn to the vertical position automatically.

2. Please use this procedure only when the network primary controller is missing or otherwise inoperable.

3. Default master pin code is "12345678", please modify it after reset:

- 1) Enter Master Mode (* + * + Master PIN Code + #)
- 2) Enter "1" to select Modify Master Pin Code.
- 3) Enter NEW 4-10 Digit Master PIN Code followed by " #"
- 4) Repeat Step 3 to confirm NEW Master PIN Code

Z-Wave™ DSK Location

You can find the QR code when you open the battery box

You may also find the QR Code and DSK on individual package of each product.

Please do not remove or damage them.

About Product

This product can be operated in any Z-Wave™ network with other Z-Wave™ certified devices from other manufacturers. All mains operated nodes within the network will act as repeaters regardless of vendor to increase reliability of the network.

About Security

This device is a security enabled Z - Wave Plus product that is able to use encrypted Z - Wave Plus messages to communicate to other security enabled Z - Wave Plus products. S2 Security Enabled Controller is required to operate the device.

SOFTWARE FUNCTION DEFINITION

User Behavior Interaction

<u>User behavior</u>	<u>Out of the network</u>	<u>In the network</u>
<u>Power on</u>	<u>keypad light is on</u> <u>The unlocking indicator light or</u> <u>locking indicator light is on</u> <u>Send Inclusion Requests for SmartStart</u>	<u>keypad light is on</u> <u>The unlocking indicator light or</u> <u>locking indicator light is on</u>
<u>Inclusion network</u>	<u>Enter master mode(* + * + Master PIN Code +</u> <u>#)</u> <u>Input "0" to enter "Network settings"</u> <u>Input "1" + "#" to enter "Inclusion mode"</u>	<u>NA</u>
<u>Exclusion network</u>	<u>NA</u>	<u>Enter master mode(* + * + Master PIN Code +</u> <u>#)</u> <u>Input "0" to enter "Network settings"</u> <u>Input "2" + "#" to enter "Exclusion mode"</u>
<u>unlock</u>	<u>The green indicator lights up for 3</u> <u>seconds</u>	<u>The green indicator lights up for 3</u> <u>seconds</u> <u>Send Notification Report and Door Lock</u> <u>Operation Report to association group</u>
<u>Lock</u>	<u>The red indicator lights up for 3</u> <u>seconds</u>	<u>The red indicator lights up for 3</u> <u>seconds</u> <u>Send Notification Report and Door Lock</u> <u>Operation Report to association group</u>
<u>User code</u> <u>(Add / delete /</u> <u>modify)</u>	<u>NA</u>	<u>Send Notification Report and User code</u> <u>Report to association group</u>

Supported Command Classes

<u>Command Class</u>	<u>Version</u>	<u>Not added</u>	<u>Non-secure added</u>	<u>Securely 0 added</u>		<u>Securely 2 added</u>	
				<u>Non-secure CC</u>	<u>Secure CC</u>	<u>Non-secure CC</u>	<u>Secure CC</u>
<u>ZWAVEPLUS_INFO</u>	<u>2</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>ASSOCIATION</u>	<u>2</u>				<u>Support</u>		<u>Support</u>
<u>ASSOCIATION_GRP_INFO</u>	<u>3</u>				<u>Support</u>		<u>Support</u>
<u>Multichannel Association</u>	<u>3</u>				<u>Support</u>		<u>Support</u>
<u>NOTIFICATION</u>	<u>8</u>				<u>Support</u>		<u>Support</u>
<u>Door Lock</u>	<u>4</u>				<u>Support</u>		<u>Support</u>
<u>User Code</u>	<u>2</u>				<u>Support</u>		<u>Support</u>
<u>Schedule Command Class</u>	<u>4</u>				<u>Support</u>		<u>Support</u>
<u>Door Lock Logging</u>	<u>1</u>				<u>Support</u>		<u>Support</u>
<u>Configuration</u>	<u>4</u>				<u>Support</u>		<u>Support</u>
<u>Application Status</u>	<u>1</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>Time</u>	<u>1</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>Time Parameters</u>	<u>1</u>				<u>Support</u>		<u>Support</u>
<u>VERSION</u>	<u>3</u>				<u>Support</u>		<u>Support</u>
<u>MANUFACTURER_SPECIFIC</u>	<u>2</u>				<u>Support</u>		<u>Support</u>
<u>DEVICE_RESET_LOCALLY</u>	<u>1</u>				<u>Support</u>		<u>Support</u>
<u>POWERLEVEL</u>	<u>1</u>				<u>Support</u>		<u>Support</u>
<u>Indicator</u>	<u>3</u>				<u>Support</u>		<u>Support</u>
<u>BATTERY</u>	<u>1</u>				<u>Support</u>		<u>Support</u>
<u>SECURITY</u>	<u>1</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>SECURITY 2</u>	<u>1</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>TRANSPORT_SERVICE</u>	<u>2</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>SUPERVISION</u>	<u>1</u>	<u>Support</u>	<u>Support</u>	<u>Support</u>		<u>Support</u>	
<u>FIRMWARE_UPDATE_MD</u>	<u>5</u>				<u>Support</u>		<u>Support</u>

Basic Command Class mapping

Basic Command maps to Door Lock Command Class, as shown below.

<u>Command</u>	<u>Value</u>	<u>Mapped</u>	<u>Value</u>	<u>Function</u>
<u>Basic Set</u>	<u>0x00</u>	<u>Door Lock Operation Set</u>	<u>Door Unsecured</u>	<u>Open the door</u>
	<u>0xFF</u>		<u>Door Secured</u>	<u>Close the door</u>
<u>Basic Report</u>	<u>0x00</u>	<u>Door Lock Operation Report</u>	<u>Door Unsecured</u>	<u>The door is opened</u>
	<u>0xFF</u>		<u>Door Secured</u>	<u>The door is closed</u>
<u>Basic Get</u>		<u>Door Lock Operation Get</u>		

ZWAVEPLUS_INFO

<u>Parameter</u>	<u>Value</u>
<u>Version</u>	<u>2</u>
<u>Role Type</u>	<u>7 (ZWAVEPLUS_INFO_REPORT_ROLE_TYPE_SLAVE_SLEEPING_LISTENING)</u> <u>(APPLICATION_FREQ_LISTENING_MODE_1000ms)</u>
<u>Node Type</u>	<u>0 (ZWAVEPLUS_INFO_REPORT_NODE_TYPE_ZWAVEPLUS_NODE)</u>
<u>Installer Icon Type</u>	<u>0x0300 (ICON_TYPE_GENERIC_DOOR_LOCK_KEYPAD)</u>
<u>User Icon Type</u>	<u>0x0300 (ICON_TYPE_GENERIC_DOOR_LOCK_KEYPAD)</u>

Manufacturer Specific

The Command is used to advertise manufacturer specific information.

<u>Parameter</u>	<u>Value</u>
<u>Manufacturer ID 1</u>	<u>0x02</u>
<u>Manufacturer ID 2</u>	<u>0x1D</u>
<u>Product Type ID 1</u>	<u>0x00 (EU) / 0x01 (US) / 0x02 (AU)</u>
<u>Product Type ID 2</u>	<u>0x03</u>
<u>Product ID 1</u>	<u>0x00</u>
<u>Product ID 2</u>	<u>0xCC (204)</u>

Version

The Command may be used to obtain the Z-Wave™ library type, the Z-Wave™ protocol version used by the application, the individual command class versions used by the application.

<u>Parameter</u>	<u>Value</u>
<u>Protocol Library Type</u>	<u>0x03</u>
<u>Protocol Version</u>	<u>0x07</u>
<u>Protocol Sub Version</u>	<u>0x12</u>
<u>Firmware 0 Version</u>	<u>Main Chip Firmware Version Major</u>
<u>Firmware 0 Sub Version</u>	<u>Main Chip Firmware Version Minor</u>
<u>Hardware Version</u>	<u>0xCC (204)</u>
<u>Number of firmware targets</u>	<u>0x01</u>
<u>Firmware 1 Version</u>	<u>Door Lock Firmware Version Major</u>
<u>Firmware 1 Sub Version</u>	<u>Door Lock Firmware Version Minor</u>

Association Group Info

The Command is used to manage associations to Node ID destinations.

<u>ID</u>	<u>Name</u>	<u>Count</u>	<u>Profile</u>	<u>Function</u>
<u>1</u>	<u>Lifeline</u>	<u>5</u>	<u>General:</u> <u>Lifeline</u> <u>(0x0001)</u>	<u>Device Reset Locally Notification (0x5A01) :</u> <u>Issued when Factory Reset is performed.</u> <u>Battery Report(0x8003):</u> <u>Issued periodically to report the current battery level;</u> <u>Issued when battery becomes low.</u> <u>Door Lock Operation Report(0x6203):</u> <u>Issued when door lock mode changed</u> <u>Notification Report(0x7105):</u> <u>Issued when door lock event changed</u> <u>User Code Report(0x6303):</u> <u>Issued when user code changed by keypad</u> <u>Master Code Report(0x6310)</u> <u>Issued when master code changed by keypad</u> <u>User Code KEYPAD MODE Report(0x630A)</u> <u>Issued when keypad mode changed by keypad</u> <u>Configuration Report(0x7006)</u> <u>Issued when the volume configuration value is changed by keypad</u> <u>Indicator Report(0x8703):</u> <u>Issued when indicator set received.</u>
<u>2</u>	<u>On/Off control</u>	<u>5</u>	<u>General: Control</u> <u>(0x2000)</u>	<u>Basic Set (0x2001) :</u> <u>Issued when door lock mode changed (value is determined by parameter 0x04)</u>

Notification

The Command is used to advertise events or states, such.

Notification Type = Access Control (0x06), according to Table 4-8-1

Notification Type = Home Security (0x07), according to Table 4-8-2

Table 4-8-1:

<u>Description</u>	<u>Event</u>	<u>Alarm type</u>	<u>Alarm Level</u>	<u>Parameters</u>
<u>Manual lock operation</u> <u>(mechanical key or thumb turn)</u>	<u>0x01</u>	<u>0x15</u>	<u>0x01</u>	<u>0x01</u>
<u>Manual unlock operation</u> <u>(mechanical key or thumb turn)</u>	<u>0x02</u>	<u>0x16</u>	<u>0x01</u>	<u>0x01</u>
<u>RF lock operation</u>	<u>0x03</u>	<u>0x18</u>	<u>0x01</u>	<u>0x01</u>
<u>RF unlock operation</u>	<u>0x04</u>	<u>0x19</u>	<u>0x01</u>	<u>0x01</u>
<u>Keypad lock operation</u>	<u>0x05</u>	<u>0x15</u>	<u>0x02</u>	
<u>Keypad unlock operation</u>	<u>0x06</u>	<u>0x13</u>	<u>User ID</u>	<u>User Code Report</u>

			<u>0xFB</u>	<u>Master Code Report</u>
<u>Auto lock locked operation</u>	<u>0x09</u>	<u>0x1B</u>	<u>0x01</u>	
<u>Lock jammed</u>	<u>0x0B</u>	<u>0x00</u>	<u>0x00</u>	
<u>All user codes deleted</u>	<u>0x0C</u>	<u>0x21</u>	<u>0xFF</u>	
<u>Single user code deleted</u>	<u>0x0D</u>	<u>0x21</u>	<u>User Id</u>	<u>User Code Report</u>
<u>New user code added (add and modify)</u>	<u>0x0E</u>	<u>0x70</u>	<u>User Id</u>	<u>User Code Report</u>
<u>New user code not added due to duplicate code</u>	<u>0x0F</u>	<u>0x71</u>	<u>User Id</u>	<u>User Code Report</u>
<u>New program code entered : unique code for lock configuration</u>	<u>0x12</u>	<u>0x70</u>	<u>0x00</u>	<u>Master Code Report</u>
<u>Manually enter user access code exceeds code limit</u>	<u>0x13</u>	<u>0x00</u>	<u>0x00</u>	

Table 4-8-2:

<u>Description</u>	<u>Notification Event</u>	<u>Alarm type</u>	<u>Alarm Level</u>	<u>Parameters</u>
<u>Tampering Product covering removed (front escutcheon removed from door)</u>	<u>0x03</u>	<u>0x00</u>	<u>0x00</u>	

Door Lock

The Command is used to operate and configure a door lock device.

Support Door Lock Mode: Door Secured (0xFF), Door Unsecured (0x00)

The auto relock time mapping table of door lock and Z-Wave™ is as follows:

<u>Z-Wave™ definition</u>	<u>Door lock function</u>
<u>Auto-relock time = 0</u>	<u>disabled</u>
<u>Auto-relock time = 1 – 30</u>	<u>30 Seconds</u>
<u>Auto-relock time = 31 – 60</u>	<u>60 Seconds</u>
<u>Auto-relock time = 61 – 120</u>	<u>120 Seconds</u>
<u>Auto-relock time = 120 – 65535</u>	<u>180 Seconds</u>

Door Lock Logging

The Command provides an audit trail in an access control application. Each time an event takes place at the door lock, the system logs the user's ID, date, time etc.

User Code

The Command is used to manage User Codes in access control systems

- Support master code
- Supported User ID Status: available(0x00), enabled/grant access (0x01), Disable(0x02)

3. Supported Keypad Modes: Normal mode, Vacation mode, Privacy mode

In Privacy mode, it will ignore Door Lock Operator Set Command, and will send Application Rejected Request.

Schedule

The Command allows scheduling the execution of commands for a given duration in a supporting device. It is a generic Command Class that may be used to schedule commands of any other Command Class.

1. SCHEDULE SUPPORTED REPORT:

- 1) Number of Supported Schedule IDs: 10
- 2) Support Enable / disable: 0
- 3) Fall-back Support: 0
- 4) Start Time Support: 1
- 5) Number of supported CC: 1
- 6) Supported CC 1: USER CODE COMMAND CLASS
- 7) Supported Command1: 0x01 (Only Set Commands are supported)
- 8) Number of Supported Schedule Blocks: 1

2. SCHEDULE SUPPORTED COMMANDS REPORT:

- 1) Supported Command Class: USER CODE COMMAND CLASS
- 2) Supported Command: USER CODE SET

Battery

The Command is used to request and report battery levels for a given device.

1. If send Battery Get to the device, it will issue Battery Report with current battery level to the requester.
2. If re-power on, it will issue Battery Report with current battery level via Lifeline.
3. If waked up, it will detect battery level, and issue Battery Report (Value=0xFF) via Lifeline when battery level is less than 20%.
4. Support timed battery report, refer to Configuration Parameter 3.

Indicator

The Command is used to help end users to monitor the operation or condition of the application provided by a supporting node.

<u>Indicator ID</u>		<u>Property ID</u>	
<u>Node Identify</u>	<u>0x50</u>	<u>On Off Period</u>	<u>0x03</u>
		<u>On Off Cycles</u>	<u>0x04</u>
		<u>On time within an On/Off period</u>	<u>0x05</u>

Time

The Command is used to read date and time from a supporting node in a Z-Wave™ network.

Time Parameters

The Command is used to set date and time in a device hosting this facility

Configuration

The Command allows product specific configuration parameters to be changed.

Note: No Bulk Support equals to True. *It will return an Application Rejected Request Command when receiving Configuration Bulk Set or Get (if received without Supervision encapsulation).* It will reset all its configuration parameters if either manually reset to factory default or receives a Configuration Default Reset Command. It will NOT modify or reset any configuration parameter when being included or excluded of a Z-Wave™ network.

<u>Parameter</u>	<u>0x01 (1)</u>			
<u>Name</u>	<u>Configure volume level</u>			
<u>Info</u>	<u>Configure volume level</u>			
<u>Properties</u>	<u>Size</u>	<u>1</u>	<u>Min Value</u>	<u>0</u>
	<u>Format</u>	<u>Unsigned Integer</u>	<u>Max Value</u>	<u>2</u>
	<u>Read-only</u>	<u>False</u>	<u>Default Value</u>	<u>2</u>
	<u>Altering capabilities</u>	<u>False</u>	<u>Advanced</u>	<u>False</u>
<u>Description</u>	<u>Configure volume level</u>			
	<u>Value</u>	<u>Function</u>		
	<u>0</u>	<u>silent</u>		
	<u>1</u>	<u>Low volume</u>		
	<u>2</u>	<u>High volume</u>		

<u>Parameter</u>	<u>0x02 (2)</u>
<u>Name</u>	<u>Configure led</u>
<u>Info</u>	<u>Enable/disable led</u>

<u>Properties</u>	<u>Size</u>	<u>1</u>	<u>Min Value</u>	<u>0</u>
	<u>Format</u>	<u>Unsigned Integer</u>	<u>Max Value</u>	<u>1</u>
	<u>Read-only</u>	<u>False</u>	<u>Default Value</u>	<u>1</u>
	<u>Altering capabilities</u>	<u>False</u>	<u>Advanced</u>	<u>False</u>
<u>Description</u>	<u>LED on the inside of the product.</u>			
	<u>Value</u>	<u>Function</u>		
	<u>0</u>	<u>OFF</u>		
	<u>1</u>	<u>ON (blink)</u>		

<u>Parameter</u>	<u>0x03 (3)</u>			
<u>Name</u>	<u>Timed battery report</u>			
<u>Info</u>	<u>Configure timed battery report</u>			
<u>Properties</u>	<u>Size</u>	<u>2</u>	<u>Min Value</u>	<u>0</u>
	<u>Format</u>	<u>Unsigned Integer</u>	<u>Max Value</u>	<u>14400 (0x3840)</u> <u>(10 days)</u>
	<u>Read-only</u>	<u>False</u>	<u>Default Value</u>	<u>1440 (0x05A0)</u> <u>(one day)</u>
	<u>Altering capabilities</u>	<u>False</u>	<u>Advanced</u>	<u>False</u>
<u>Description</u>	<u>Enable/Disable timed battery report and sets how often battery is reported in minutes.</u>			
	<u>Value</u>	<u>Function</u>		
	<u>0</u>	<u>Disable timed battery report</u>		
	<u>1-14400</u>	<u>Enable timed battery report and sets how often battery is reported in minutes.</u>		

<u>Parameter</u>	<u>0x04 (4)</u>			
<u>Name</u>	<u>Association Group 2 Settings</u>			
<u>Info</u>	<u>Configure the Basic Set value</u>			

<u>Properties</u>	<u>Size</u>	<u>1</u>	<u>Min Value</u>	<u>0</u>
	<u>Format</u>	<u>Unsigned Integer</u>	<u>Max Value</u>	<u>6</u>
	<u>Read-only</u>	<u>False</u>	<u>Default Value</u>	<u>1</u>
	<u>Altering capabilities</u>	<u>False</u>	<u>Advanced</u>	<u>False</u>
<u>Description</u>	<u>Determine the Basic Set value to control other Z-Wave™ devices directly.</u>			
	<u>Value</u>	<u>Function</u>		
	<u>0</u>	<u>Disable completely.</u>		
	<u>1</u>	<u>Unlocked Status sends Basic SET 0xFF, Locked Status sends Basic SET 0x00.</u>		
	<u>2</u>	<u>Unlocked Status sends Basic SET 0x00, Locked Status sends Basic Set 0xFF.</u>		
	<u>3</u>	<u>Sends only Basic SET 0xFF when Unlocked Status happens.</u>		
	<u>4</u>	<u>Sends only Basic SET 0x00 when Locked Status happens.</u>		
	<u>5</u>	<u>Sends only Basic SET 0x00 when Unlocked Status happens.</u>		
	<u>6</u>	<u>Sends only Basic SET 0xFF when Locked Status happens.</u>		